



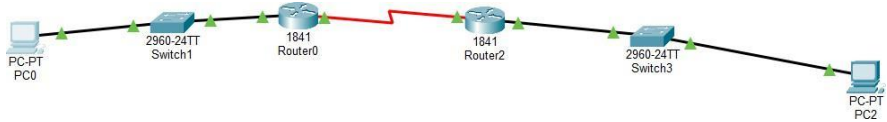
BHARATIYA VIDYA BHAVAN'S
SARDAR PATEL INSTITUTE OF TECHNOLOGY
Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai – 400058-India
Department of Computer Engineering

Name	Siddh Sakariya
UID no.	2024300200
Experiment No.	9

AIM:	Experiment on Cisco Packet Tracer
Program 1	
PROBLEM STATEMENT:	Task 1: Subnet the Address Space. Task 2: Determine Interface Addresses. Task 3: Configure the Serial and FastEthernet Addresses. Task 4: Verify the Configurations. Task 5: Reflection.
THEORY:	What is Cisco Packet Tracer? Cisco Packet Tracer is a network simulation software developed by Cisco Systems. It allows users to create, configure, and troubleshoot network topologies in a virtual environment. It is widely used by students and professionals to understand networking concepts without requiring physical hardware. It provides a drag-and-drop interface to add routers, switches, PCs, and connections. Users can configure devices using CLI or GUI, similar to real networking devices. What is Network Topology? Network topology refers to the physical or logical arrangement of devices in a network. It defines how devices are connected and how data flows between them. Types include: Star, Bus, Ring, and Mesh. In this experiment, we use a simple routed topology with two routers and two LANs. What is Subnetting? Subnetting is the process of dividing a large network into smaller networks (subnets). It improves network performance, security, and efficient IP usage. Static Routing



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	Static routing is a method where routes are manually configured in routers. Without routing, devices on separate networks cannot communicate. Static routes ensure proper packet forwarding between routers
TASKS:	 Task 1: Subnet the Address Space The given address space 192.168.1.0/24 was subnetted into 3 subnets using a /27 prefix (255.255.255.224), each supporting up to 30 usable hosts. Step 1: Network Requirements • Network connected to R1 requires addresses for 15 hosts. • Network connected to R2 requires addresses for 30 hosts. • The WAN link between R1 and R2 requires IP addresses at each end. Step 2: Subnet Design Answers • Number of subnets needed: 3 • Subnet mask (dotted decimal): 255.255.255.224 • Subnet mask (slash format): /27 • Usable hosts per subnet: 30 Step 3: Subnet Assignments • Subnet 1 → Network attached to R1: 192.168.1.32/27 • Subnet 2 → WAN link between R1 and R2: 192.168.1.64/27 • Subnet 3 → Network attached to R2: 192.168.1.96/27 Task 2: Determine Interface Addresses Address Assignment • R1 LAN interface (Fa0/0): 192.168.1.33 (first valid host in Subnet 1)



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- PC0 (on R1 LAN): 192.168.1.34, Gateway: 192.168.1.33
- R1 WAN interface (Serial0/1/0): 192.168.1.65 (first valid host in Subnet 2)
- R2 WAN interface (Serial0/1/0): 192.168.1.66 (second valid host in Subnet 2)
- R2 LAN interface (Fa0/0): 192.168.1.97 (first valid host in Subnet 3)
- PC2 (on R2 LAN): 192.168.1.98, Gateway: 192.168.1.97

Task 3: Configure Serial and FastEthernet Addresses

The following screenshots show the router and PC interface configurations as implemented in Cisco Packet Tracer.

Router0 (R1) – FastEthernet0/0 Configuration

IP Address: 192.168.1.33 | Subnet Mask: 255.255.255.224 | Port Status: On

The screenshot shows the configuration window for Router0, specifically the configuration for the FastEthernet0/0 interface. The window has tabs for Physical, Config, CLI, and Attributes. The Config tab is active, and the left sidebar shows a tree view with categories: GLOBAL, Settings, Algorithm Settings, ROUTING, Static, RIP, SWITCHING, VLAN Database, INTERFACE, and a list of interfaces including FastEthernet0/0, FastEthernet0/1, Serial0/1/0, and Serial0/1/1. The main area shows the configuration for FastEthernet0/0 with the following settings: Port Status is On, Link Speed is 100 Mbps, Duplex is Full Duplex, MAC Address is 000C.CFB3.6001, IP Configuration is set to 192.168.1.33 with Subnet Mask 255.255.255.224, and Tx Ring Limit is 10.

Router0 (R1) – Serial0/1/0 Configuration

IP Address: 192.168.1.65 | Subnet Mask: 255.255.255.224 | Clock Rate: 64000



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The screenshot shows the configuration window for Router0, specifically for the Serial0/1/0 interface. The left sidebar contains a tree view with categories: GLOBAL, Settings, Algorithm Settings, ROUTING, Static, RIP, SWITCHING, VLAN Database, INTERFACE, FastEthernet0/0, FastEthernet0/1, Serial0/1/0 (selected), and Serial0/1/1. The main area displays the configuration for Serial0/1/0 with the following settings:

Serial0/1/0	
Port Status	☒ On
Duplex	☒ Full Duplex
Clock Rate	64000
IP Configuration	
IPv4 Address	192.168.1.65
Subnet Mask	255.255.255.224
Tx Ring Limit	10

Router2 (R2) – FastEthernet0/0 Configuration

IP Address: 192.168.1.97 | Subnet Mask: 255.255.255.224 | Port Status: On

The screenshot shows the configuration window for Router2, specifically for the FastEthernet0/0 interface. The left sidebar contains a tree view with categories: GLOBAL, Settings, Algorithm Settings, ROUTING, Static, RIP, SWITCHING, VLAN Database, INTERFACE, FastEthernet0/0 (selected), FastEthernet0/1, Serial0/1/0, and Serial0/1/1. The main area displays the configuration for FastEthernet0/0 with the following settings:

FastEthernet0/0	
Port Status	☒ On
Link Speed	☒ 100 Mbps ☐ 10 Mbps ☒ Auto
Duplex	☐ Half Duplex ☒ Full Duplex ☒ Auto
MAC Address	0006.2A7D.8701
IP Configuration	
IPv4 Address	192.168.1.97
Subnet Mask	255.255.255.224
Tx Ring Limit	10

Router2 (R2) – Serial0/1/0 Configuration

IP Address: 192.168.1.66 | Subnet Mask: 255.255.255.224 | Clock Rate:



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1200

Router2

Physical Config CLI Attributes

GLOBAL

- Settings
- Algorithm Settings
- ROUTING**
- Static
- RIP
- SWITCHING**
- VLAN Database
- INTERFACE**
- FastEthernet0/0
- FastEthernet0/1
- Serial0/1/0
- Serial0/1/1

Serial0/1/0

Port Status ☒ On

Duplex ☒ Full Duplex

Clock Rate 1200

IP Configuration

IPv4 Address 192.168.1.66

Subnet Mask 255.255.255.224

Tx Ring Limit 10

PC0 – IP Configuration

IP: 192.168.1.34 | Subnet Mask: 255.255.255.224 | Default Gateway:
192.168.1.33

PC0

Physical Config Desktop Programming Attributes

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.1.34

Subnet Mask 255.255.255.224

Default Gateway 192.168.1.33

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::240:BFF:FEE1:1E70

Default Gateway

DNS Server



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PC0

Physical Config Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

Global Settings

Display Name PC0

Interfaces FastEthernet0

Gateway/DNS IPv4

☐ DHCP

☒ Static

Default Gateway 192.168.1.33

DNS Server

Gateway/DNS IPv6

☐ Automatic

☒ Static

Default Gateway

DNS Server

Device Clock: 01:01:59 Mon Mar 1 1993 UTC

NTP

PC2 – IP Configuration

IP: 192.168.1.98 | Subnet Mask: 255.255.255.224 | Default Gateway:
192.168.1.97

PC2

Physical Config Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Link Speed ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 000A.F30B.2644

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 192.168.1.98

Subnet Mask 255.255.255.0

IPv6 Configuration

☐ Automatic

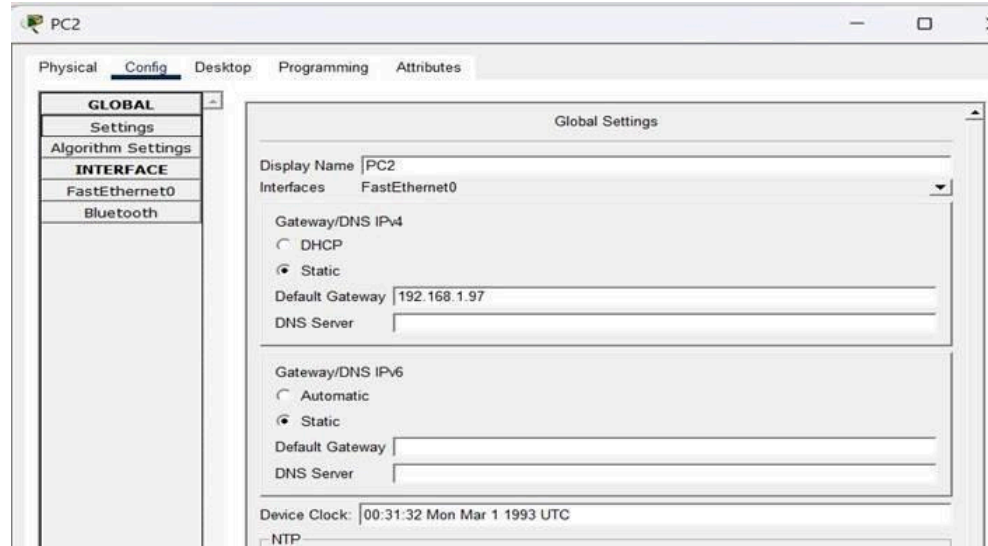
☒ Static

IPv6 Address FE80::20A:F3FF:FE0B:2644

Link Local Address FE80::20A:F3FF:FE0B:2644



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This shows the configuration of router interfaces including FastEthernet and Serial ports. Each interface is assigned an IP address and subnet mask based on the addressing scheme. The interfaces are enabled to allow data transmission between networks. Proper router configuration ensures connectivity between different subnets. This step is essential for establishing network communication.

Task 4: Verify the Configurations

The following ping results verify that the network is operating correctly after static routing was configured.

Ping Results from PC0 (R0's Host)



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```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.97

Pinging 192.168.1.97 with 32 bytes of data:

Reply from 192.168.1.97: bytes=32 time=9ms TTL=254
Reply from 192.168.1.97: bytes=32 time=8ms TTL=254
Reply from 192.168.1.97: bytes=32 time=1ms TTL=254
Reply from 192.168.1.97: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.1.97:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 9ms, Average = 4ms
C:\>ping 192.168.1.98

Pinging 192.168.1.98 with 32 bytes of data:

Reply from 192.168.1.98: bytes=32 time=1ms TTL=126
Reply from 192.168.1.98: bytes=32 time=1ms TTL=126
Reply from 192.168.1.98: bytes=32 time=13ms TTL=126
Reply from 192.168.1.98: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.1.98:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 13ms, Average = 4ms
C:\>
```

Image 2: PC0 pinging R2 gateway (192.168.1.97) and PC2 (192.168.1.98) – 0% loss

Ping Results from PC2 (R2's Host)

```
PC2
Physical Config Desktop Programming Attributes
Command Prompt

Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 5ms, Average = 3ms
C:\>ping 192.168.1.33

Pinging 192.168.1.33 with 32 bytes of data:

Reply from 192.168.1.33: bytes=32 time=2ms TTL=254
Reply from 192.168.1.33: bytes=32 time=1ms TTL=254
Reply from 192.168.1.33: bytes=32 time=13ms TTL=254
Reply from 192.168.1.33: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.1.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 13ms, Average = 4ms
C:\>ping 192.168.1.34

Pinging 192.168.1.34 with 32 bytes of data:

Reply from 192.168.1.34: bytes=32 time=1ms TTL=126
Reply from 192.168.1.34: bytes=32 time=1ms TTL=126
Reply from 192.168.1.34: bytes=32 time=1ms TTL=126
Reply from 192.168.1.34: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>ping 192.168.1.34

Pinging 192.168.1.34 with 32 bytes of data:

Reply from 192.168.1.34: bytes=32 time=1ms TTL=126
Reply from 192.168.1.34: bytes=32 time=1ms TTL=126
Reply from 192.168.1.34: bytes=32 time=1ms TTL=126
Reply from 192.168.1.34: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.1.34:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\>
```




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Verification Answers

Q1. From the host attached to R1, is it possible to ping the default gateway?

Yes. PC0 (192.168.1.34) successfully pinged its default gateway 192.168.1.33 (R1's Fa0/0 interface) with 0% packet loss, as confirmed in the ping results shown in Image 2. All 4 packets were sent and received with TTL=255, confirming direct LAN connectivity.

Q2. From the host attached to R2, is it possible to ping the default gateway?

Yes. PC2 (192.168.1.98) successfully pinged its default gateway 192.168.1.97 (R2's Fa0/0 interface). The ping results shown in Image 1 confirm 0% packet loss with replies showing TTL=254, indicating direct LAN-level communication.

Q3. From the host attached to R1, is it possible to ping the host on R2's network?

Yes. PC0 successfully pinged PC2 (192.168.1.98) across the routers. The ping in Image 2 shows all 4 packets received with TTL=126, which confirms that the packet traversed two routers (R1 and R2) using static routing. This validates cross-network communication.

Q4. From the host attached to R2, is it possible to ping the host on R1's network?

Yes. PC2 successfully pinged PC0 (192.168.1.34) with 0% packet loss. The ping results in Image 1 show TTL=126, confirming bi-directional inter-network communication through static routing between R1 and R2.

Task 5: Reflection

Q1. Are there any devices on the network that cannot ping each other?

Initially, before static routing was configured, devices on different subnets



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	(PC0 on Subnet 1 and PC2 on Subnet 3) could not communicate with each other. The routers had no knowledge of how to forward packets to remote subnets. After configuring static routes on Router0 using the command <code>ip route 192.168.1.96 255.255.255.224 192.168.1.66</code>, all devices in the network were able to ping each other successfully with 0% packet loss. The ping screenshots in Images 1 and 2 confirm full connectivity across all devices. Q2. What is missing from the network that is preventing communication between devices? Static routing was initially missing from the network configuration. Without routing entries, routers do not know how to forward packets destined for remote networks. Even though all interfaces were correctly configured with valid IP addresses, the absence of static routes meant packets from PC0 trying to reach PC2 (and vice versa) had no valid path through the routers. Once static routing was configured on both routers defining next-hop addresses through the WAN serial link (192.168.1.64/27 subnet), full end-to-end communication was established between all devices across the two LANs.
CONCLUSION:	In this experiment, a network was successfully created and configured using Cisco Packet Tracer. The given address space 192.168.1.0/24 was subnetted into three smaller /27 networks, and appropriate IP addresses were assigned to all devices including routers (R1, R2) and PCs (PC0, PC2). Router interfaces — both FastEthernet (LAN) and Serial (WAN) — were configured and activated properly using the Cisco Config GUI and CLI. Static routing was implemented to enable communication between the different subnets. The connectivity was verified using ping commands, confirming that all devices could communicate properly across the network. The TTL values observed in ping results (255 for same-subnet, 254 for one hop, 126 for two hops) validated the correct routing path across routers. This experiment provided hands-on understanding of subnetting, IP addressing, router interface configuration, and static routing concepts